

Student Academic Performance Prediction

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SECTION 1

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1. Abstract

This project aims to predict student academic performance categories using a dataset of 3,000 records and 32 features. The workflow includes data ingestion, checks on data structures, and addressing missing values using median and mode imputation. Categorical features are converted into numerical data with label encoding, and all inputs are normalized through min-max scaling.

A Decision Tree Classifier was used for multi-class prediction, achieving a training and test accuracy of 1.0. The analysis shows that the classification is entirely driven by the `'final_exam_score'` feature, indicating a data leakage problem.

The framework illustrates key pre-processing steps and model evaluation techniques for educational data analytics.

2. Introduction

This project uses Python and Scikit-Learn to create a machine learning pipeline that analyzes student behaviors, demographics, and grades.

I received training in several essential topics:

- Running code in Jupyter Notebook and Google Colab.
- Data manipulation and structural profiling with Pandas and NumPy.
- Identifying and handling missing data fields through median and mode imputation.
- Feature engineering, specifically using `'LabelEncoder'` and `'MinMaxScaler'`.
- Building, training, and evaluating supervised machine learning models with `'DecisionTreeClassifier'`.

3. Project Objective

- To create an end-to-end data pre-processing and machine learning pipeline for student records.
- To effectively address missing data using appropriate imputation methods.
- To convert mixed-type text and numeric attributes into standard format numbers for ML models.
- To train a Decision Tree Classifier to accurately predict a student's final academic performance category.
- To evaluate model metrics and identify key factors that drive student success.

4. Methodology

The project follows a structured workflow for data preparation and modeling:

- **Data Collection:** Loaded `'student_performance_dataset.csv'` (with 3,000 rows and 32 columns) into a Pandas DataFrame.
- **Data Profiling:** Identified 26 numerical columns and 6 categorical text columns through structural inspection tools (`.info()`).
- **Data Cleaning:** Found 60 missing data fields across `'social_media_hours'`, `'focus_score'`, and `'sleep_hours'`. Numerical gaps were filled with the column median, and categorical gaps with the column mode.

- **Feature Engineering:** Used a 'LabelEncoder' to convert all text columns into unique integers.
- **Data Splitting:** Separated the target variable ('performance_category') from the features, creating an 80% training set (2,400 rows) and a 20% testing set (600 rows) with a fixed 'random_state=42'.
- **Data Scaling:** Applied 'MinMaxScaler' to normalize all input features to a scale between 0.0 and 1.0.
- **Model Training:** Initialized and trained a 'DecisionTreeClassifier' using the processed training data.

Code Repository Link: <https://github.com/divyaagr/Student-Academic-Performance-Prediction-IDEAS-.git>

5. Data Analysis and Results

➔ Missing Data Treatment

Missing data was limited to three lifestyle features and was resolved during pre-processing:

Feature Name	Missing Count (Before)	Missing Count (After)	Imputation Method
social_media_hours	60	0	Column Median

Feature Name	Missing Count (Before)	Missing Count (After)	Imputation Method
focus_score	60	0	Column Median
sleep_hours	60	0	Column Median

➔ Model Performance & Feature Importance

The Decision Tree Classifier evaluated on the dataset provided the following metrics:

- **Training Set Accuracy:** 1.0 (100%)
- **Testing Set Accuracy:** 1.0 (100%)

Top Ranked Feature	Feature Importance Score	Remaining 31 Features	Feature Importance Score
final_exam_score	1.0000	study_hours, attendance_percentage, etc.	0.0000

Analysis Note: The model's 1.0 accuracy stems from its complete reliance on `final\_exam\_score` for determining category boundaries.

This indicates data leakage, as the target variable clearly depends on this score, making other behavioral inputs unnecessary for the model.

6. Conclusion

The project completed a full machine learning pipeline, from raw data ingestion to feature scaling and model training. The Decision Tree model achieved a perfect accuracy score of 1.0. However, the feature importance results indicate it ignores lifestyle metrics like study hours and sleep patterns because the final exam score is included in the input data. This is a clear case of data leakage.

Recommendations for Future Work:

- Remove the `final\_exam\_score` column from the input variables to ensure the model predicts academic performance categories based only on behavioral, environmental, and mental traits.
- After removing the exam score, test and compare different algorithms—such as Random Forest or Logistic Regression—alongside the Decision Tree to see which best handles behavioral features.